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Production Review · Workshop, Chiang Rai 20 kWp

20 kW three-phase hybrid inverter, no battery yet, zero export · July–October 2026 (October to the 30th) · Data: SolisCloud · Prepared 30 October 2026 · Rev 04

1,226 kWh
Solar production, October 1–30

2.04
kWh/kWp per day in October (limited)

3.5–4.3
kWh/kWp per day when export was allowed (22–28 Oct)

600–1,000
kWh/month of solar lost to zero-export limiting

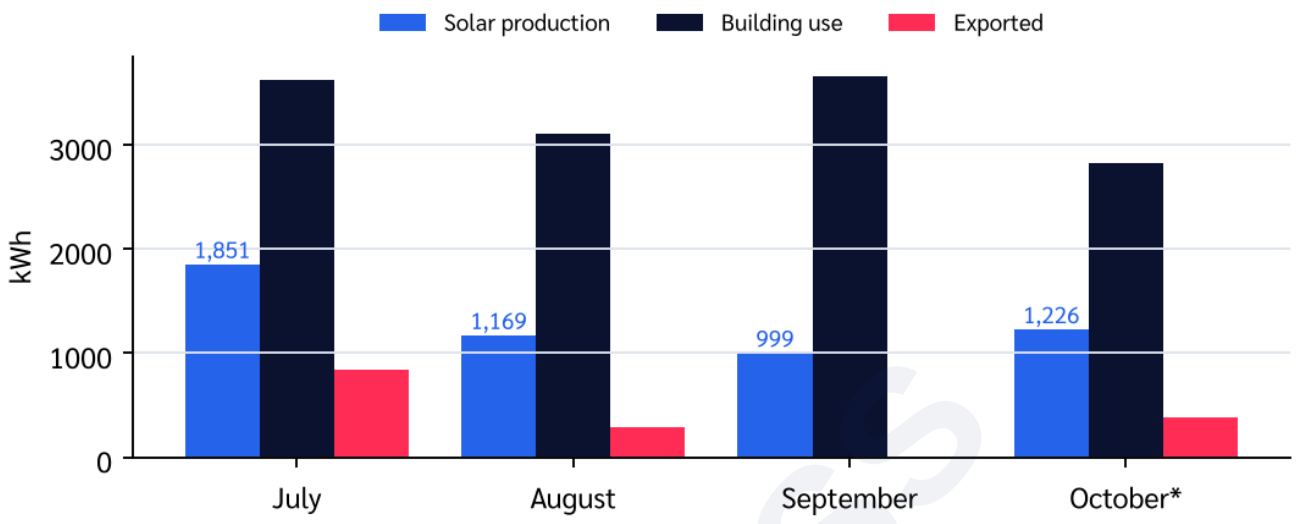
Summary

- The panels and inverter are healthy. On days when the inverter could run freely (22–28 October) the plant made 69–86 kWh/day, or 3.5–4.3 kWh/kWp, which is normal output for 20 kWp in Chiang Rai.
- Production is low because of the zero-export setting: the inverter reduces its output to whatever the building uses at that moment. On Sundays (3–5 kWh load) it produces almost nothing.
- July was the best month (1,851 kWh) because export was allowed. After export was disabled in August, production fell by about a third and has stayed between 1,000 and 1,230 kWh/month.
- The building uses the same amount of solar every month, about 840–1,000 kWh (28–30% of its consumption). Solar it cannot use at that moment is either exported or lost.
- September grid figures are not reliable because the main 3-phase meter was wired with the wrong phases until 30 September. Production figures come from the inverter and are reliable.
- **Phase imbalance:** on normal days phase L1 carries about 58% of the building load, L2 only 12% and L3 30%, which is about 80% imbalance. The inverter follows each phase separately, but L1 reaches its per-phase limit of about 29 A while spare solar on L2 and L3 cannot be used under zero export (see Phase balance).

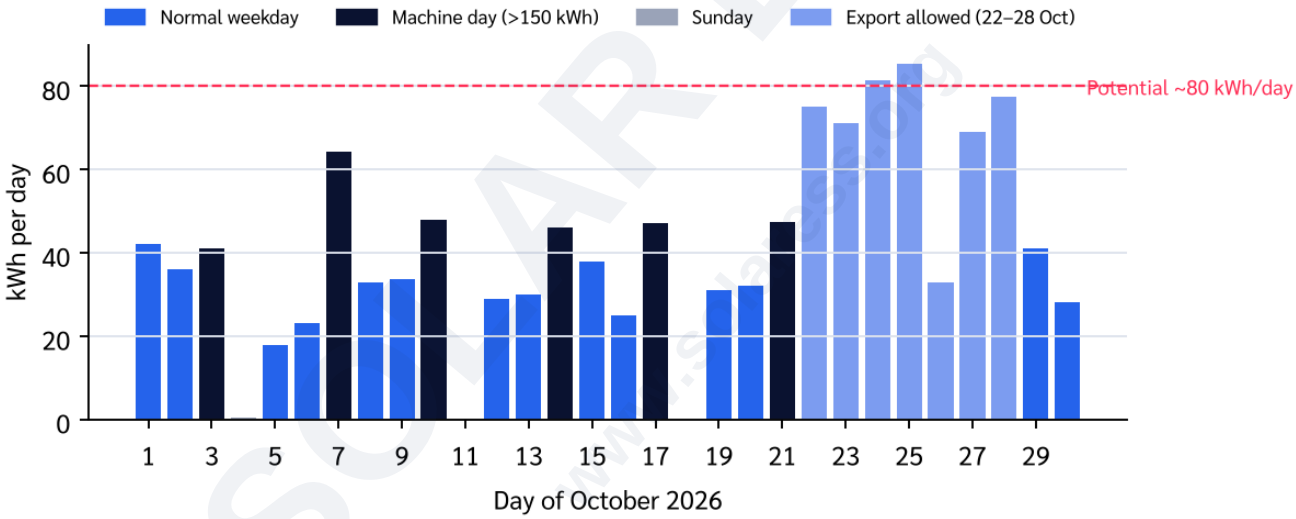
Monthly comparison: July to October 2026

Month	Solar kWh	kWh/day	kWh/kWp/day	Use kWh	Bought kWh	Exported kWh	Solar used kWh	Solar share
July	**1,851**	59.7	2.99	3,619	2,456	844	1,007	28%
August	**1,169**	37.7	1.89	3,106	2,026	283	886	29%
September	**999**	33.3	1.67	3,659	4,689 ⚠	2,899 ⚠	–	–
October*	**1,226**	40.9	2.04	2,828	1,844	388	838	30%

* October covers 1–30 October. ⚠ September grid import and export are distorted by the phase-wiring fault (export exceeds production), so solar used on site is not shown. Solar used = production − export. Solar share = solar used ÷ building use.



October 2026 by day



Day type	Days	Solar kWh/day	kWh/kWp	Use kWh/day	Export kWh/day
Normal weekday	14	31.4	1.57	48.6	3.0
Machine day (>150 kWh)	6	48.9	2.45	223.7	7.9
Sunday	3	0.1	0.01	3.9	0.6
Export allowed (22–28 Oct)	7	70.4	3.52	102.6	42.3

Findings

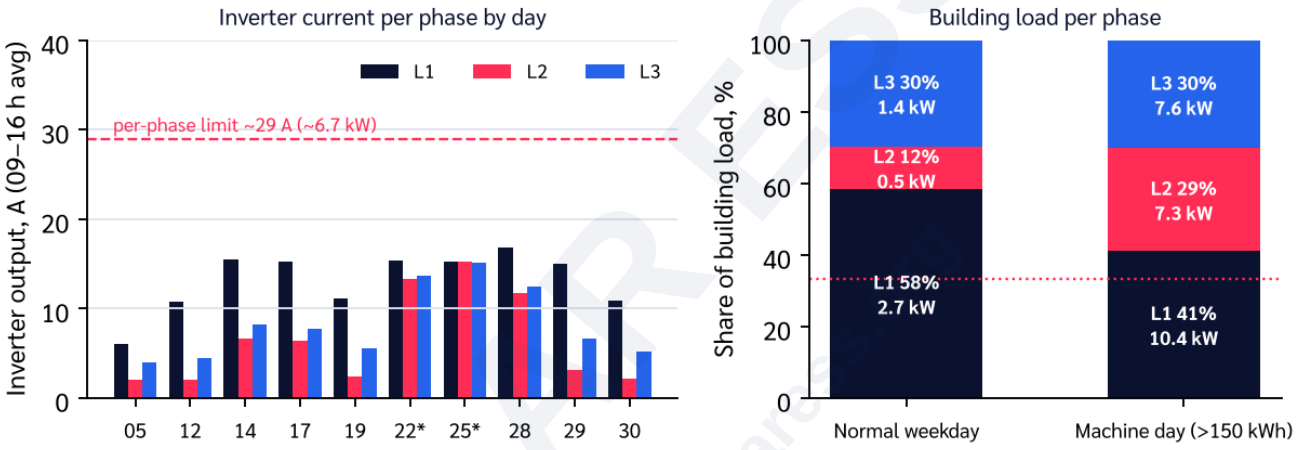
1. Normal weekdays: the inverter follows the building load and makes only about 1.5 kWh/kWp, even on clear days.
2. Machine days (every 3–4 days, 180–265 kWh): output rises because the load is higher, but the building still buys 130–213 kWh from PEA on each of these days, because most of the machine load does not fall in sunny hours.
3. Sundays: load is only 3–5 kWh, so the inverter almost shuts down (0–0.4 kWh on 4, 11 and 18 October).
4. 22–28 October: export appears to have been allowed. Output rose to 69–86 kWh/day and 296 kWh went to the grid. From 29 October output fell back, which suggests export was limited again.

5. No equipment fault is visible in the data. The difference between months is explained by the export setting, the wiring fault in September and the rainy season.

> **Lost production:** compared with what the plant made when export was allowed, roughly **600–1,000 kWh/month** of solar is not produced under zero export. This is an estimate based on 30 days of data.

Phase balance (load imbalance between L1, L2 and L3)

The site has a three-phase supply and the Solis S6-EH3P20K-H, a 20 kW three-phase high-voltage hybrid inverter. We compared the inverter output current on each phase with the main meter reading on each phase for 10 days in October, using the 09:00–16:00 average. The building load on each phase is estimated as the inverter output plus the power bought from the grid on that phase.



Left: average inverter current per phase; * = days when export was allowed, so output is equal on all phases. Right: average building load per phase; the dotted line is a perfectly balanced 33%.

Date	Day type	L1 A	L2 A	L3 A	Load L1 kW	Load L2 kW	Load L3 kW	Imbalance	Max A
05/10	Normal weekday	6.0	2.0	4.0	1.7	0.5	1.3	**72%**	19
12/10	Normal weekday	10.8	2.1	4.5	2.5	0.5	1.2	**81%**	19
14/10	Machine day (>150 kWh)	15.5	6.6	8.2	10.3	7.1	7.5	**31%**	30
17/10	Machine day (>150 kWh)	15.2	6.4	7.7	11.2	8.1	8.4	**27%**	29
19/10	Normal weekday	11.2	2.4	5.6	2.7	0.5	1.4	**79%**	19
22/10	Export allowed (22–28 Oct)	15.4	13.3	13.7	2.5	0.4	1.3	**86%**	23
25/10	Export allowed (22–28 Oct)	15.3	15.2	15.1		load near zero (Sunday)		–	22
28/10	Machine day (>150 kWh)	16.8	11.8	12.5	9.8	6.5	7.0	**33%**	29
29/10	Normal weekday	15.0	3.1	6.6	3.9	0.7	1.6	**82%**	29
30/10	Normal weekday	10.9	2.2	5.2	2.6	0.5	1.4	**80%**	20

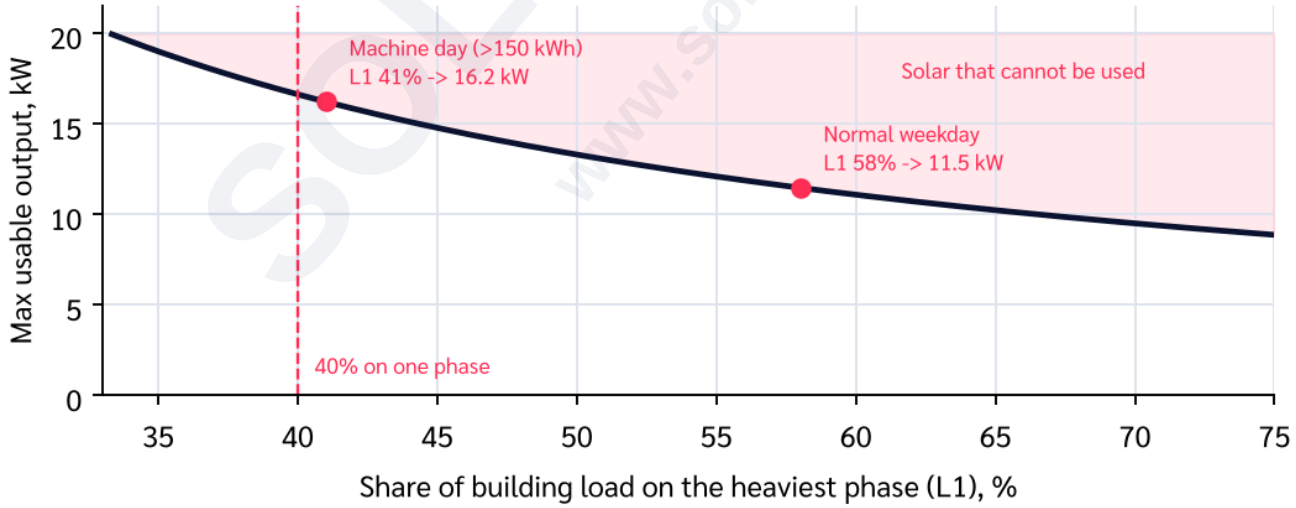
Imbalance = (highest phase − lowest phase) ÷ highest phase. kW is estimated at 230 V without power factor, so it is approximate. Max A is the highest 5-minute inverter current on any phase that day.

What the inverter datasheet says (Solis S6-EH3P20K-H)

These are the figures in the Solis S6-EH3P(12–20)K-H datasheet that decide how the 20 kW model behaves with an unbalanced load.

Datasheet item	S6-EH3P20K-H	What it means at this site
Rated AC output power	**20 kW**	The total across all three phases.
Max output current per phase	**28.9 A (230/400 V)** **30.4 A (220/380 V)**	About 6.65 kW per phase at 230 V, which is one-third of 20 kW. L1 peaked at 27.6–30 A at the site, the same as this limit.
Max allowed phase unbalance (grid and backup)	**100%**	The inverter is designed to keep running with any imbalance. It does not shut down or trip, and the October data show no trips.
Max power per phase (grid and backup)	**40–50% of rated**	The series lists both values. At 230 V the current limit above is reached first.
Max usable PV input	**32 kW, 4 MPPT**	The panels are not the bottleneck.
Battery port	**120–800 V, max 20 kW**	A high-voltage battery can store solar that no phase can use.

Why output falls when imbalance is over 40%. Under zero export the inverter can give each phase only what that phase is using, and no more than 28.9 A (about 6.65 kW) on any phase. The most solar the building can use is therefore the sum, across the three phases, of either that phase's load or 6.65 kW, whichever is smaller. With a balanced load the inverter can use all 20 kW. When one phase carries more than its third, that phase reaches 6.65 kW first and buys the rest from the grid. Meanwhile the lightly loaded phases have no load to take more solar. Once L1 carries 40% of the load, the most the inverter can usefully deliver falls to 16.6 kW. At this site's 58% on normal days it falls to 11.5 kW.



Load on heaviest phase	Max usable output	% of 20 kW	Note
Balanced (33%)	20.0 kW	100%	Full rating usable
40%	16.6 kW	83%	The per-phase power figure in the datasheet
41% (site, machine day)	16.2 kW	81%	
50%	13.3 kW	66%	
58% (site, normal day)	11.5 kW	57%	L1 limits the whole system

Seen at the site, 29 October 13:00–14:20: L1 ran at 26.1–27.7 A, at the 28.9 A limit. L2 ran at 0–0.1 A and L3 at 0.4–0.6 A. Total output was only 5.9–6.1 kW, and at 14:18 L1 was still buying 0.99 kW from the grid. With export allowed the same panels made 11–14 kW (22–28 October). The inverter did not trip: L1 reached its per-phase limit and the remaining solar had nowhere to go.

Phase-balance findings

1. **The load is strongly unbalanced:** 72–86% on normal days, well over 40%. Phase L1 carries 2.5–3.9 kW, L2 only about 0.5 kW and L3 about 1.3 kW. On machine days the imbalance drops to 27–33% because the machines are three-phase.
2. **The inverter is already following each phase separately.** Grid power stays close to zero on every phase. One phase exports while another imports in only 0–11% of intervals. This shows the "Unbalanced Output" setting is effectively ON: if it were OFF, the output would be equal on all phases, L2 would export, and zero-export control would cut the whole inverter back further.
3. **L1 reaches the per-phase ceiling.** The highest L1 current is 29–30 A, about 6.7 kW, which is one-third of 20 kW. On 29 October L1 hit this ceiling and bought an average of 0.45 kW from the grid while L2 and L3 had unused solar. Zero export does not allow that surplus to be sent out, so the inverter cannot compensate for it. This costs roughly 2–3 kWh on heavy-L1 days.
4. **When export is allowed, the output becomes balanced.** On 25 October the inverter delivered 15.1–15.3 A on every phase (about 11 kW). This confirms the hardware can deliver on all three phases.
5. **Imbalance is a secondary loss.** The main loss is still zero export against a small daytime load of about 4 kW, which cuts back about 6 kW of output (see Findings). Rebalancing removes the L1 bottleneck. It does not by itself recover the curtailed energy; a battery, an export permit or more daytime load is needed for that.

In short: with a load more than 40% unbalanced and zero export, a three-phase inverter can only give each phase what that phase uses, up to about one-third of its rating. The heavily loaded phase (L1) runs out of headroom and buys from the grid, and the lightly loaded phase (L2) leaves solar unused. Moving about 1–1.5 kW of daytime load from L1 to L2 is the lowest-cost fix.

Recommendations

1. **Shift flexible machine work to 10:00–15:00 on sunny days** so solar meets the load directly. No cost.
2. **Add a battery** to store midday surplus for the evening and for machine days. The 22–28 October data shows how much surplus is available.
3. **Apply to PEA to sell surplus power** (2.20 baht/kWh) so the inverter can run fully without a compliance risk.
4. **Confirm and record the export setting** (it appears to have changed on about 22 and 29 October) and keep it consistent while a decision is made.
5. **Rebalance the single-phase loads:** move about 1–1.5 kW of daytime single-phase load (e.g. an air conditioner, water heater or machine) from L1 to L2 at the distribution board, and keep the inverter setting "Unbalanced Output" ON (Grid Feed-in Power Limit – EPM).

Appendix: daily data, 1–30 October 2026

Date	Day	Solar kWh	kWh/kWp	Use kWh	Bought kWh	Exported kWh	Day type
01/10	Thu	42.0	2.10	47.0	13.8	12.1	Normal weekday
02/10	Fri	36.0	1.80	57.0	20.4	1.7	Normal weekday

Date	Day	Solar kWh	kWh/kWp	Use kWh	Bought kWh	Exported kWh	Day type
03/10	Sat	41.0	2.05	180.4	138.4	2.2	Machine day (>150 kWh)
04/10	Sun	0.4	0.02	3.2	3.4	0.6	Sunday
05/10	Mon	17.9	0.90	29.6	10.6	0.9	Normal weekday
06/10	Tue	23.0	1.15	42.8	18.9	1.4	Normal weekday
07/10	Wed	64.3	3.22	181.4	130.4	19.6	Machine day (>150 kWh)
08/10	Thu	32.9	1.65	38.9	13.6	11.4	Normal weekday
09/10	Fri	33.8	1.69	48.7	14.6	1.8	Normal weekday
10/10	Sat	48.0	2.40	235.5	181.9	6.8	Machine day (>150 kWh)
11/10	Sun	0.0	0.00	4.0	3.9	0.6	Sunday
12/10	Mon	29.0	1.45	38.4	8.6	1.4	Normal weekday
13/10	Tue	30.0	1.50	51.4	20.4	1.5	Normal weekday
14/10	Wed	46.0	2.30	237.0	186.4	5.5	Machine day (>150 kWh)
15/10	Thu	38.0	1.90	62.6	23.6	1.2	Normal weekday
16/10	Fri	25.0	1.25	46.2	20.1	1.3	Normal weekday
17/10	Sat	47.0	2.35	266.0	213.2	7.9	Machine day (>150 kWh)
18/10	Sun	0.0	0.00	4.6	4.8	0.6	Sunday
19/10	Mon	31.0	1.55	41.6	10.2	1.6	Normal weekday
20/10	Tue	32.0	1.60	68.8	35.8	1.5	Normal weekday
21/10	Wed	47.3	2.37	241.9	189.1	5.2	Machine day (>150 kWh)
22/10	Thu	75.1	3.76	52.7	22.1	46.6	Export allowed (22–28 Oct)
23/10	Fri	71.1	3.56	55.5	27.9	45.7	Export allowed (22–28 Oct)
24/10	Sat	81.4	4.07	256.0	182.0	28.3	Export allowed (22–28 Oct)
25/10	Sun	85.5	4.28	5.0	4.4	87.3	Export allowed (22–28 Oct)
26/10	Mon	33.0	1.65	45.3	28.1	18.6	Export allowed (22–28 Oct)
27/10	Tue	69.0	3.45	41.2	12.9	42.7	Export allowed (22–28 Oct)
28/10	Wed	77.5	3.88	262.8	187.9	27.0	Export allowed (22–28 Oct)
29/10	Thu	41.0	2.05	63.7	23.1	3.4	Normal weekday
30/10	Fri	28.2	1.41	43.3	14.1	1.3	Normal weekday

Source: SolisCloud monitoring, plant data downloaded 30 October 2026, 20:45. Potential output reference: 20 kWp × about 4 kWh/kWp per clear day in Chiang Rai. Figures are estimates, not a guarantee.

Phase-balance references: Solis, "Zero export solutions – balancing three-phase loads" (average, minimum and unbalanced-load methods); Solis S5/S6 hybrid zero-export setting (Unbalanced Output ON/OFF); Hopewind, "Zero output under unbalanced load". Per-phase data: SolisCloud inverter and meter curves, 10 days in October 2026.

Datasheet: Solis S6-EH3P(12–20)K-H, V2.2 (11/2024) and V1.8 (10/2024), values for the S6-EH3P20K-H. Site per-phase readings: SolisCloud, 29 October 2026.